

2. Global Conformal invariance

全局共形不变性

1. 共形群

共形变换: $x \rightarrow x'$ 保角 (两向量夹角不变) $\Rightarrow$ 旋转及其扩张

$$f'^{\mu} = f^{\mu} + \epsilon^{\mu} \partial_{\rho} f^{\rho}$$

$$\text{保角: } f'_{\mu} f'_{\nu} g^{\mu\nu} = f_{\mu} f_{\nu} g^{\mu\nu} \Rightarrow g'_{\mu\nu} = g_{\mu\nu} - (\partial_{\mu} \epsilon_{\nu} + \partial_{\nu} \epsilon_{\mu})$$

$$g'_{\mu\nu}(x') = \Lambda(x) g_{\mu\nu}(x) \Rightarrow \text{无穷小变换: } x'^{\mu} = x^{\mu} + \epsilon^{\mu}(x), \quad g'_{\mu\nu} = g_{\mu\nu} - (\partial_{\mu} \epsilon_{\nu} + \partial_{\nu} \epsilon_{\mu}) \quad \theta'_{\mu\nu} = g_{\mu\nu} - (\partial_{\mu} \epsilon_{\nu} + \partial_{\nu} \epsilon_{\mu})$$

$$\text{元: } \Lambda(x) \equiv 1$$

$$\text{取 } f(x), \partial_{\mu} \epsilon_{\nu} + \partial_{\nu} \epsilon_{\mu} = f(x) g_{\mu\nu}, \quad f(x) = \frac{2}{d} \partial_{\rho} \epsilon^{\rho}$$

对于平直空间: $\eta_{\mu\nu}$ 对称, 对称性: $\eta_{\mu\nu} = \eta_{\nu\mu}$, $\partial_{\mu} \epsilon_{\nu} = \partial_{\nu} \epsilon_{\mu}$, ($\eta = \text{diag}[1, \dots, 1]$)

$$2\partial_{\mu} \partial_{\nu} \epsilon_{\rho} = \eta_{\mu\rho} \partial_{\nu} f + \eta_{\nu\rho} \partial_{\mu} f - \eta_{\mu\nu} \partial_{\rho} f \xrightarrow{\eta^{\mu\nu}} 2\partial^2 \epsilon_{\mu} = (2-d) \partial_{\mu} f \xrightarrow{\partial_{\nu}} 2\partial^2 \epsilon_{\nu} \partial_{\mu} = (2-d) \partial_{\mu} \partial_{\nu} f = \eta_{\mu\nu} \partial^2 f \xrightarrow{\partial^2 f} 2(d-1) \partial^2 f = 0$$

$d=1$: 无约束, 任-光滑变换保角

$d=2$: 主要研究对象

$$\partial_{\mu} \partial_{\nu} \epsilon_{\rho} = \frac{1}{2} (\eta_{\mu\rho} \partial_{\nu} B_{\rho} + \eta_{\nu\rho} \partial_{\mu} B_{\rho} - \eta_{\mu\nu} \partial_{\rho} B_{\rho}) \rightarrow C_{\mu\nu\rho} = C_{\mu\rho\nu}$$

$$d \geq 3: \partial_{\mu} \partial_{\nu} f = 0 \Rightarrow f(x) = A + B_{\mu} x^{\mu} \Rightarrow \partial^2 \epsilon_{\mu} = (1 - \frac{d}{2}) B_{\mu} \Rightarrow \underline{\epsilon_{\mu} = a_{\mu} + b_{\mu\nu} x^{\nu} + C_{\mu\nu\rho} x^{\nu} x^{\rho}}$$

$$\downarrow a_{\mu}: \text{无约束} \quad b_{\mu\nu}: \partial_{\mu} \epsilon_{\nu} + \partial_{\nu} \epsilon_{\mu} = b_{\mu\nu} + b_{\nu\mu} = \frac{2}{d} b^{\lambda} \eta_{\lambda\mu} \\ \downarrow \text{对称} \quad \text{反对称} \\ b_{\mu\nu} = \alpha \eta_{\mu\nu} + m_{\mu\nu}, \quad m_{\mu\nu} + m_{\nu\mu} = 0 \\ \text{旋转+放缩}$$

$$C_{\mu\nu\rho}: C_{\mu\nu\rho} + \eta_{\mu\nu} C_{\rho}^{\lambda} \partial_{\lambda} = \frac{1}{2} (\eta_{\mu\rho} B_{\nu} + \eta_{\nu\rho} B_{\mu} - \eta_{\mu\nu} B_{\rho})$$

$$C_{\mu\nu\rho} = \eta_{\mu\rho} b_{\nu} + \eta_{\mu\nu} b_{\rho} - \eta_{\nu\rho} b_{\mu}, \quad b_{\mu} = \frac{1}{d} C^{\lambda}{}_{\lambda} \partial_{\mu}$$

$$\hookrightarrow x'^{\mu} = x^{\mu} + 2(x \cdot b) x^{\mu} - b^{\mu} x^2 \rightarrow \frac{x'^{\mu}}{x'^2} = \frac{x^{\mu}}{x^2} - b^{\mu} + 2 \frac{(x \cdot b) x^{\mu}}{x^2} \\ = \frac{x^{\mu}}{x^2} - b^{\mu} + 2 \frac{x^{\mu} (x'_{\mu} - x_{\mu})}{x^2}$$

$$\text{变换: } \begin{array}{ll} \text{translation} & x'^{\mu} = x^{\mu} + a^{\mu} \\ \text{dilation} & x'^{\mu} = \alpha x^{\mu} \\ \text{rigid rotation} & x'^{\mu} = M^{\mu}{}_{\nu} x^{\nu} \\ \text{special conformal transition (SCT)} & x'^{\mu} = \frac{x^{\mu} - b^{\mu} x^2}{1 - 2b \cdot x + b^2 x^2} \end{array}$$

$$\Lambda(x) = (1 - 2b \cdot x + b^2 x^2)^2, \quad \frac{x'^{\mu}}{x'^2} = \frac{x^{\mu}}{x^2} - b^{\mu}$$

故SCT变换包含 $x^{\mu} \rightarrow \frac{x^{\mu}}{x^2}$, 反演

生成元

$\Rightarrow$

$$P_{\mu} = -i \partial_{\mu}$$

$$D = -i x_{\mu} \partial^{\mu}$$

$$L_{\mu\nu} = i(x_{\mu} \partial_{\nu} - x_{\nu} \partial_{\mu})$$

$$K_{\mu} = -i(2x_{\mu} x^{\nu} \partial_{\nu} - x^2 \partial_{\mu})$$

$$\text{RP: } \frac{x'^{\mu}}{x'^2} = \frac{x^{\mu}}{x^2} - b^{\mu}$$

$$\epsilon_{\mu} = i a^{\mu} P_{\mu} = i D \theta$$

$$= i \theta M^{\mu\nu} L_{\mu\nu} = i b^{\mu} K_{\mu}$$

$$(K_{\mu}: x'^{\mu} - x^{\mu} = -b^{\mu} x^2 + 2x_{\mu} b^{\mu})$$

生成元的对易关系:

$$[D, P_\mu] = i P_\mu$$

$$[K_\rho, L_{\mu\nu}] = i(\eta_{\rho\mu} K_\nu - \eta_{\rho\nu} K_\mu)$$

$d=2$ 时同构于 $SO(3,1)$, 为闵氏空间

$$[D, K_\mu] = -i K_\mu$$

$$[P_\rho, L_{\mu\nu}] = i(\eta_{\rho\mu} P_\nu - \eta_{\rho\nu} P_\mu)$$

$$[K_\mu, P_\nu] = 2i(\eta_{\mu\nu} D - L_{\mu\nu}) \quad [L_{\mu\nu}, L_{\rho\sigma}] = i(\eta_{\nu\rho} L_{\mu\sigma} + \eta_{\mu\sigma} L_{\nu\rho} - \eta_{\mu\rho} L_{\nu\sigma} - \eta_{\nu\sigma} L_{\mu\rho})$$

共形算符: $J_{\mu\nu} = L_{\mu\nu}$ $J_{-1\mu} = \frac{1}{2}(P_\mu - K_\mu)$ $J_{-10} = D$ $J_{0\mu} = \frac{1}{2}(P_\mu + K_\mu)$ 构成反对易算符

$$\begin{pmatrix} 0 & D & -P+K \\ D & 0 & -P-K \\ P & P+K & L \end{pmatrix}$$

$J_{ab} = -J_{ba} \Rightarrow$ 同构于 $SO(d+1, 1)$

$$[J_{ab}, J_{cd}] = i(\eta_{ad} J_{bc} + \eta_{bc} J_{ad} - \eta_{ac} J_{bd} - \eta_{bd} J_{ac}) \quad \eta = \text{diag}[-1, 1, \dots, 1] \quad \text{参数: } M_{\mu\nu}: \frac{d(d-1)}{2}, b_\mu: d, a: 1, a^\mu: d$$

不变性: $\Gamma(x_1 \dots x_d)$ 不变

$P_\mu, L_{\mu\nu}$: Γ 依赖于 $|x_i - x_j|$

$$\text{某记 } \frac{d+3d+2}{2} = \frac{(d+1)(d+2)}{2}$$

$$D: \Gamma \propto \frac{|x_i - x_j|}{|x_j - x_k|}, \quad K_\mu: \Gamma \propto \frac{|x_{23}|}{|x_{13}|} \frac{|x_{34}|}{|x_{24}|} \quad (\text{交叉比/非谐比})$$

2. 共形群表示

d 维共形群表示 (对经典场): 无穷小变换: $\Phi'(x) = (1 - i\omega_\mu T_\mu) \Phi(x)$

带自旋: 对 $x=0$: $L_{\mu\nu} \Phi(0) = S_{\mu\nu} \Phi(0)$ 对任意 x : $e^{ix^\rho P_\rho} L_{\mu\nu} e^{-ix^\rho P_\rho} \Phi(0) = S_{\mu\nu} - x^\rho [\eta_{\rho\mu} P_\nu - \eta_{\rho\nu} P_\mu] \Phi(0) + 0$

$$[L_{\mu\nu}, P_\rho] = -i(\eta_{\rho\mu} P_\nu - \eta_{\rho\nu} P_\mu)$$

$$= S_{\mu\nu} - x_\mu P_\nu + x_\nu P_\mu$$

故可写作: $P_\mu \Phi(x) = -i \partial_\mu \Phi(x)$

$$L_{\mu\nu} \Phi(x) = [S_{\mu\nu} + i(x_\mu \partial_\nu - x_\nu \partial_\mu)] \Phi(x)$$

同理: $S_{\mu\nu}, \tilde{D}, K_\mu$ 为 L, D, K 的 0 处表示

$$[\tilde{D}, K_\mu] = -i K_\mu$$

$$\Rightarrow e^{ix^\rho P_\rho} D e^{-ix^\rho P_\rho} = D + x^\rho P_\rho \rightarrow D \Phi(x) = (\tilde{D} - ix^\nu \partial_\nu) \Phi(x)$$

$$[\tilde{D}, S_{\mu\nu}] = 0$$

$$e^{ix^\rho P_\rho} K_\mu e^{-ix^\rho P_\rho} = K_\mu + 2x_\mu D - 2x^\nu L_{\mu\nu} + 2x_\mu (x^\rho P_\rho) - x^2 P_\mu$$

$$[K_\rho, S_{\mu\nu}] = i(\eta_{\rho\mu} K_\nu - \eta_{\rho\nu} K_\mu)$$

$$\hookrightarrow K_\mu \Phi = [K_\mu + 2x_\mu \tilde{D} - x^\nu S_{\mu\nu} - 2ix_\mu x^\nu \partial_\nu + ix^2 \partial_\mu] \Phi$$

$$[K_\mu, K_\nu] = 0$$

$$[S_{\mu\nu}, S_{\rho\sigma}] = i(\eta_{\nu\rho} S_{\mu\sigma} + \eta_{\mu\sigma} S_{\nu\rho} - \eta_{\mu\rho} S_{\nu\sigma} - \eta_{\nu\sigma} S_{\mu\rho})$$

舒尔引理: 2 个不可约表示的映射 $\delta \psi(g) = \psi(g) \Rightarrow$ 若 $\delta \psi(g) = \psi(g) \delta$

1) ψ 与 ψ 等价 ($V=W, \psi=\psi \Rightarrow \delta = \lambda I$) 2) 不等价: $\delta = 0$

等价情况: 与 $S_{\mu\nu}$ 对易的算符为 $\lambda I \Rightarrow \hat{\Delta} = -i\Delta \Rightarrow \kappa_\mu = 0$

$$\varepsilon = iD\theta = -i\theta(-i\Delta - i\kappa^\mu \partial_\mu) = \theta(\Delta + \kappa^\mu \partial_\mu)$$

$$\text{故 } \phi(x') = \phi + \varepsilon\phi = (1 - \theta(\Delta + \kappa^\mu \partial_\mu))\phi = (e^\theta)^{-\Delta}\phi(x) \Rightarrow \phi(x) = \left|\frac{\partial x'}{\partial x}\right|^{\frac{d}{2}} \phi(x')$$

标量场: $x' = \left|\frac{\partial x'}{\partial x}\right|x \Rightarrow \phi(x') = \lambda^{-d}\phi$, Δ 为标度维度 $\Rightarrow \phi(x) = \left|\frac{\partial x'}{\partial x}\right|^{-\frac{d}{2}} \phi(x')$, $g'_{\mu\nu} \Rightarrow \left|\frac{\partial x'}{\partial x}\right| = \Lambda(x)^{\frac{d}{2}} \rightarrow$ 拟自然的

能动张量: $\delta S = \int d^d x T^{\mu\nu} \partial_\mu \varepsilon_\nu$, $x'_\mu = x_\mu + \varepsilon_\mu$

$$= \frac{1}{2} \int d^d x T^{\mu\nu} (\partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu) \rightarrow \text{取 } T \text{ 为对称张量, 共形下变换: } f(x) = \frac{2}{d} \partial_\rho x^\rho \rightarrow \partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu = f(x) \eta_{\mu\nu}$$

$$= \frac{1}{2} \int d^d x T^\mu{}_\mu \partial_\rho \varepsilon^\rho \quad T^\mu{}_\mu = 0 \quad (\text{共形不变性 } \delta S = 0)$$

一般情况: $x'^\mu = (1 + \kappa) x^\mu$, $\mathcal{F}(\phi) = \phi' = (1 - \alpha \Delta) \phi$

$$\begin{aligned} j_0^\mu &= -\mathcal{L} x^\mu + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} x^\nu \partial_\nu \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi \\ &= T_\nu{}^\mu x^\nu + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi \end{aligned} \quad , \quad \partial_\mu j_0^\mu = T_\nu{}^\mu{}_{,\mu} + \Delta \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \phi \right) = 0$$

$$V^\mu = \frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} (\eta^{\mu\rho} \partial_\rho \phi + i S^{\mu\rho} \phi) \quad , \quad \text{设 } V^\mu = \partial_\alpha \sigma^{\alpha\mu} \Rightarrow \partial_\alpha^\mu = \frac{1}{2} (\delta^{\mu\alpha} + \sigma^{\mu\alpha})$$

$$\chi^{\lambda\rho\mu\nu} = \frac{2}{d-2} \left[\eta^{\lambda\rho} \partial_\alpha^{\mu\nu} - \eta^{\lambda\mu} \partial_\alpha^{\rho\nu} - \eta^{\lambda\mu} \partial_\alpha^{\nu\rho} + \eta^{\mu\nu} \partial_\alpha^{\lambda\rho} + \frac{1}{d-1} (\eta^{\lambda\rho} \eta^{\mu\nu} - \eta^{\lambda\mu} \eta^{\rho\nu}) \partial_\alpha^\alpha \right]$$

$$\partial_\alpha^{\mu\nu} \text{ 对称: } \partial_\lambda \partial_\rho \chi^{\lambda\rho\mu\nu} = \frac{2}{d-2} \left[\partial^\mu \partial_\lambda^\nu + \eta^{\mu\nu} \partial_\lambda \partial_\rho \partial_\alpha^{\lambda\rho} - 2 \partial^\mu \partial_\rho \partial_\alpha^{\rho\nu} + \frac{1}{d-1} (\partial^\mu \eta^{\mu\nu} \partial_\alpha^\alpha - \partial^\mu \partial_\alpha^\nu \partial_\alpha^\alpha) \right]$$

$$\partial_\lambda \partial_\rho \chi^{\lambda\rho\mu}{}_\mu = \frac{2}{d-2} \left[\partial^\mu \partial_\lambda^\mu + d \partial_\lambda \partial_\rho \partial_\alpha^{\lambda\rho} - 2 \partial_\lambda \partial_\rho \partial_\alpha^\nu + \frac{1}{d-1} (\partial^\mu d \partial_\alpha^\mu - \partial^\mu \partial_\alpha^\mu) \right] = 2 \partial_\rho \partial_\lambda \partial_\alpha^{\rho\lambda}$$

$$\begin{aligned} \partial_\lambda \partial_\rho \partial_\mu \chi^{\lambda\rho\mu\nu} &= \frac{2}{d-2} \left[\partial^\nu \partial_\lambda \partial_\alpha^{\mu\nu} + \partial^\nu \partial_\lambda \partial_\rho \partial_\alpha^{\lambda\rho} - 2 \partial^\nu \partial_\rho \partial_\alpha^{\rho\nu} + \frac{1}{d-1} (\partial^\nu \partial_\alpha^\nu \partial_\alpha^\alpha - \partial^\nu \partial_\alpha^\nu \partial_\alpha^\alpha) \right] = \frac{2}{d-2} \left[\partial^\nu \partial_\lambda \partial_\rho \partial_\alpha^{\lambda\rho} - \partial^\nu \partial_\rho \partial_\alpha^{\rho\nu} + \frac{1}{d-1} (0) \right] \\ &= \frac{2}{d-2} \left[\partial^\nu \partial_\lambda V^\lambda - \partial^\nu V^\nu \right] = 0 \end{aligned}$$

$$\chi^{\lambda\rho\mu\nu}{}_{[\mu\nu]} = \frac{1}{2} (\chi^{\lambda\rho\mu\nu} - \chi^{\lambda\rho\nu\mu}) = \frac{\partial_\alpha^\alpha}{(d+1)(d-2)} (\eta^{\lambda\mu} \eta^{\rho\nu} - \eta^{\lambda\nu} \eta^{\rho\mu}) ,$$

$$\text{故: } \frac{1}{2} \partial_\lambda \partial_\rho \chi^{\lambda\rho\mu}{}_\mu = \partial_\mu V^\mu; \quad \partial_\rho B^{\rho\mu}{}_\mu = \frac{1}{2} i \partial_\rho \left(\frac{\delta \mathcal{L}}{\delta (\partial_\rho \phi)} S^{\mu\rho} \phi \right)$$

$$T^{\mu\nu} = T_\nu{}^\mu + \partial_\rho B^{\rho\mu\nu} + \frac{1}{2} \partial_\lambda \partial_\rho \chi^{\lambda\rho\mu\nu} \Rightarrow T^\mu{}_\mu = \partial_\mu j_0^\mu \rightarrow j_0^\mu = T^\mu{}_\nu x^\nu$$

3. 传播子:

2点关联函数: $\langle \phi(x_1) \phi(x_2) \rangle = \frac{1}{Z} \int D\phi \phi(x_1) \phi(x_2) e^{-S[\phi]}$, ϕ_1, ϕ_2 可为不同场

坐标变换: $\langle \phi(x_1) \phi(x_2) \rangle = \left| \frac{\partial x'}{\partial x} \right|_{x=x_1}^{\Delta_1} \left| \frac{\partial x'}{\partial x} \right|_{x=x_2}^{\Delta_2} \langle \phi(x'_1) \phi(x'_2) \rangle$

→ dilation: $x \rightarrow \lambda x$:

$\langle \phi(x_1) \phi(x_2) \rangle = \lambda^{\Delta_1 + \Delta_2} \langle \phi(x'_1) \phi(x'_2) \rangle \rightarrow$ Poincaré 对称: $\langle \phi(x_1) \phi(x_2) \rangle = f(|x_1 - x_2|)$

故 $\langle \phi(x_1) \phi(x_2) \rangle = \frac{C_{12}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}}$

→ SCT: $\left| \frac{\partial x'}{\partial x} \right| = \frac{1}{(1 - 2bx + b^2x^2)^d} \Rightarrow \frac{C_{12}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}} = \frac{C_{12}}{y_1^{\Delta_1} y_2^{\Delta_2}} \frac{(y_1 y_2)^{\frac{\Delta_1 + \Delta_2}{2}}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}}, y_i = (1 - 2bx_i + b^2x_i^2)$

代入: $\Delta_1 \neq \Delta_2$ 时上式为0, 即: $\langle \phi(x_1) \phi(x_2) \rangle = \begin{cases} \frac{C_{12}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}} & \Delta_1 = \Delta_2 \\ 0 & \text{else} \end{cases}$

是何物?

三点函数: $\langle \phi(x_1) \phi(x_2) \phi(x_3) \rangle = \frac{C_{123}}{x_{12}^a x_{23}^b x_{13}^c} = \frac{C_{123}}{x_{12}^a x_{23}^b x_{13}^c} \frac{(y_1 y_2)^{\frac{a}{2}} (y_2 y_3)^{\frac{b}{2}} (y_1 y_3)^{\frac{c}{2}}}{y_1^{\Delta_1} y_2^{\Delta_2} y_3^{\Delta_3}} \quad \text{即: } \begin{cases} a = \Delta_1 + \Delta_2 - \Delta_3 \\ b = \Delta_2 + \Delta_3 - \Delta_1 \\ c = \Delta_1 + \Delta_3 - \Delta_2 \end{cases}$

故: $\langle \phi(x_1) \phi(x_2) \phi(x_3) \rangle = \frac{C_{123}}{x_{12}^{\Delta_1 + \Delta_2 - \Delta_3} x_{23}^{\Delta_2 + \Delta_3 - \Delta_1} x_{13}^{\Delta_1 + \Delta_3 - \Delta_2}}$

多体:

可构造标形不变量: 非简比: $f(\frac{x_{12} x_{34}}{x_{13} x_{24}}, \dots) \Rightarrow \langle \phi_1 \dots \phi_n \rangle = f(\dots) \prod_{i,j} x_{ij}^{a_{ij}}$

Wald 恒等式: $\partial_\mu \langle T^{\mu\nu} X \rangle = - \sum \delta(x - x_i) \frac{\partial}{\partial x_i} \nu \langle X \rangle$, $X = \prod_{i=1}^n x_i$

→ Lorentz 对称性:

↓ $j^{\mu\nu\rho} = T^{\mu\nu} x^\rho - T^{\mu\rho} x^\nu$

↓ $\partial_\mu \langle (T^{\mu\nu} x^\rho - T^{\mu\rho} x^\nu) X \rangle = \sum \delta(x - x_i) [(x_i^\nu \partial_i^\rho - x_i^\rho \partial_i^\nu) \langle X \rangle - i S^{\nu\rho} \langle X \rangle]$

莱布尼兹律, 代入

$$\hookrightarrow \text{代} \lambda: \langle (T^{\mu\nu} - T^{\nu\mu}) X \rangle = -i\tilde{\epsilon} \delta(x-x_i) S^{\nu\mu} \langle x \rangle \quad (\text{for Lorentz})$$

$$\text{对 } D = -ix^\nu \partial_\nu \Delta \rightarrow \partial_\mu \langle T^\mu{}_\nu X^\nu X \rangle = -\sum \delta(x-x_i) (x^\nu \partial_i \nu \langle x \rangle + \Delta_i \langle X \rangle)$$

$$\hookrightarrow \langle T^\mu{}_\mu X \rangle = -\sum \delta(x-x_i) \Delta_i \langle X \rangle \quad (\text{for dilation})$$

能动张量关联:

$$S_{\mu\nu\rho\sigma}(x) = \langle T_{\mu\nu}(x) T_{\rho\sigma}(0) \rangle \xrightarrow{\text{坐标}} (\mu, \nu); (\rho, \sigma)$$

$$\downarrow \xrightarrow{\text{平移不变}} S_{\mu\nu\rho\sigma}(x) = \langle T_{\mu\nu}(0) T_{\rho\sigma}(-x) \rangle = \langle T_{\rho\sigma}(-x) T_{\mu\nu}(0) \rangle = S_{\rho\sigma\mu\nu}(-x) \xrightarrow{\text{宇称守恒}} S_{\mu\nu\rho\sigma} = S_{\rho\sigma\mu\nu}$$

$$\downarrow \xrightarrow{\text{dilation}} x \rightarrow \lambda x: S_{\mu\nu\rho\sigma}(\lambda x) = \lambda^{-4} S_{\mu\nu\rho\sigma}(x) \rightarrow S \propto x^{-4} \quad (d=2)$$

$$S_{\mu\nu\rho\sigma} = (x^2)^{-4} \left(A_1 g_{\mu\nu} g_{\rho\sigma} (x^2)^2 + A_2 (g_{\mu\rho} g_{\nu\sigma} + g_{\mu\sigma} g_{\nu\rho}) (x^2)^2 \right. \\ \left. + A_3 (g_{\mu\nu} x_\rho x_\sigma + g_{\rho\sigma} x_\mu x_\nu) x^2 + A_4 x_\mu x_\nu x_\rho x_\sigma \right)$$

$$\text{约束: 守恒流: } \partial_\mu T^{\mu\nu} = 0 \Rightarrow \partial^\mu S_{\mu\nu\rho\sigma} = 0$$

$$\downarrow \partial^\mu S_{\mu\nu\rho\sigma}(x) = -(x^2)^{-4} \left(3(A_4 + 2A_3) x_\nu x_\rho x_\sigma + (4A_1 + 3A_2) g_{\rho\sigma} x_\nu x^2 + (4A_2 - A_3)(g_{\rho\nu} x_\sigma + g_{\nu\sigma} x_\rho) x^2 \right)$$

$$\downarrow \neq 0: A_1 = 3A; A_2 = -A; A_3 = -4A; A_4 = 8A$$

$$S_{\mu\nu\rho\sigma}(x) = \frac{A}{(x^2)^4} \left((3g_{\mu\nu} g_{\rho\sigma} - g_{\mu\rho} g_{\nu\sigma} - g_{\mu\sigma} g_{\nu\rho}) (x^2)^2 - 4x^2 (g_{\mu\nu} x_\rho x_\sigma + g_{\rho\sigma} x_\mu x_\nu) + 8x_\mu x_\nu x_\rho x_\sigma \right)$$

$$S^\mu{}_\mu{}^\sigma{}_\sigma = \langle T^\mu{}_\mu(x) T^\sigma{}_\sigma(0) \rangle = \frac{A}{(x^2)^4} (x^2)^2 \cdot (3d^2 - 10d + 8) = \frac{A}{(x^2)^2} ((d-2)(2d-4))$$

$$\text{故 } d=2 \text{ 时 } \langle T^\mu{}_\mu(x) T^\sigma{}_\sigma(0) \rangle = 0$$

$$\hookrightarrow \langle T^\mu{}_\mu(0)^2 \rangle = 0 \rightarrow \text{Traceless}$$